StreamPlus

STREAMPLUS PREMIUM-TIER UPGRADE TARGETING

Optimizing Subscriber Offers to Maximize Revenue and Minimize Churn



EXECUTIVE SUMMARY

Goal: Increase ARPU by upgrading Basic → Premium

Upside: +$96 annual revenue per successful upgrade

Risk: Wrong targeting → subscriber churn

Cost of mistake: ~$55 LTV per false positive

Key challenge: False positives are costly → need precision over volume

Objective: Build a predictive model to identify likely upgraders

Focus: Maximize net value, minimize churn risk

Takeaway: Targeted approach = higher conversions + fewer losses

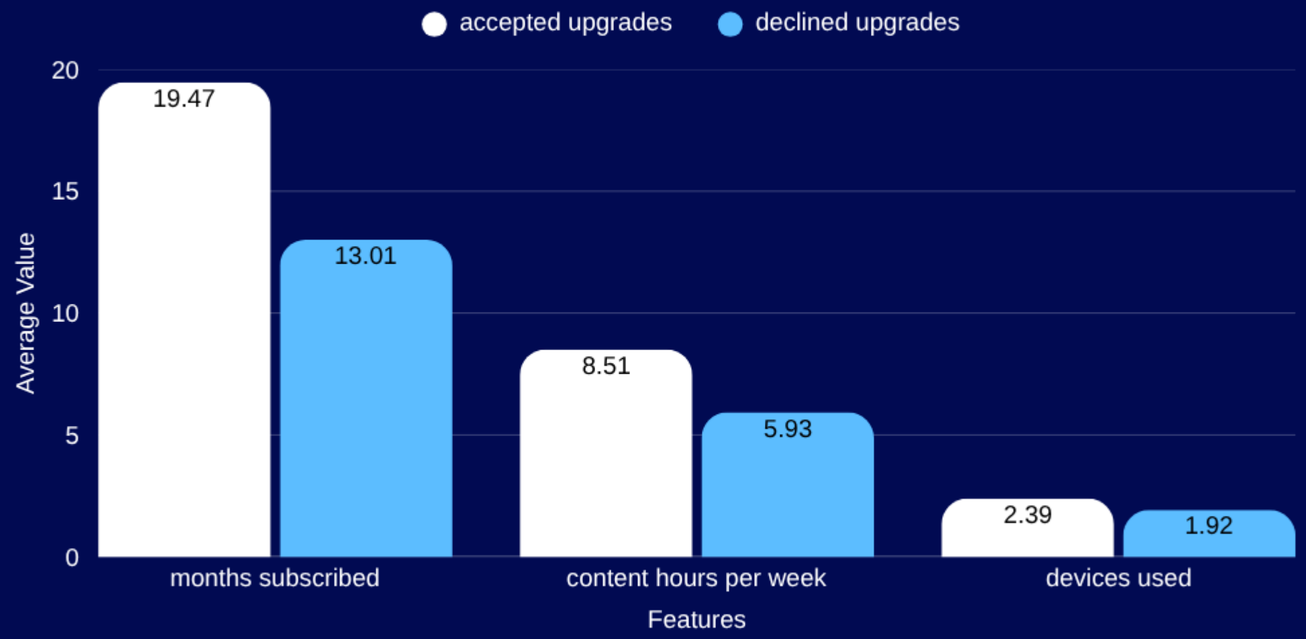


THE BUSINESS PROBLEM

- **Sending upgrade offers** to everyone can backfire
- Trying to **increase revenue** → can cause churn
- **False Positive (FP):** wrong person gets offer → may cancel → **$55 loss**
- **False Negative (FN):** miss potential upgrader → lose **$96 opportunity** *(no added risk)*
- **Key takeaway:** Target the right customers › reach more people

DATA EXPLORATION





2,500 Subscribers Analyzed

31.08% Upgrade Acceptance

No Missing Values

SUBSCRIBER BEHAVIOR INSIGHTS



Engagement Metrics

Longer tenure and higher weekly viewing hours indicate engaged subscribers ready to upgrade.



Multi-Device Users

Users with 3-4 connected devices strongly skew towards upgrade acceptance.



Our analysis reveals clear behavioral patterns that predict upgrade likelihood. **Longer subscription tenure** strongly correlates with higher upgrade rates, indicating that loyal subscribers are more receptive to premium offers.

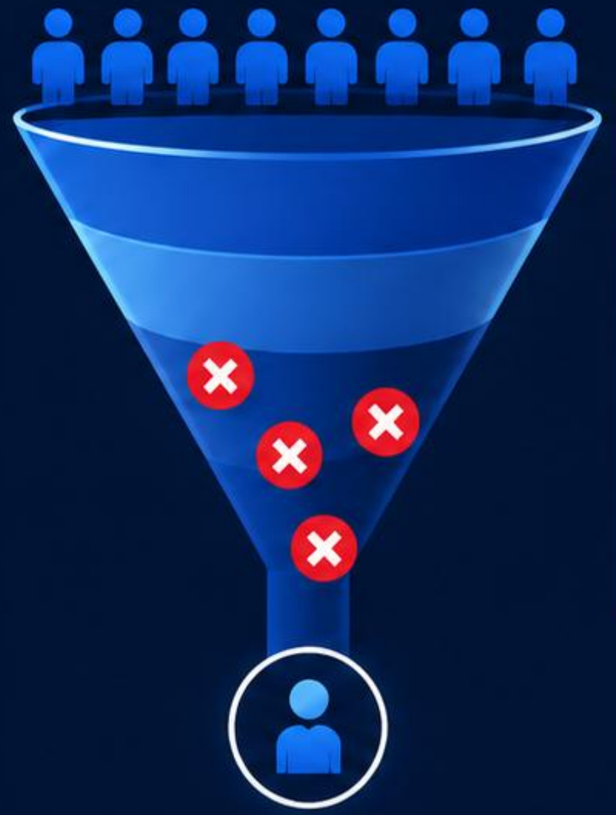
Additionally, users consuming **more content hours per week** demonstrate higher engagement and are significantly more likely to accept upgrades. **Younger subscribers** also show slightly higher upgrade propensity, suggesting age-based targeting opportunities.

MODEL BUILDING & INTERPRETATION

LOGISTIC REGRESSION MODEL

- Negative = Predicted No Upgrade
  Positive = Predicted Upgrade
- Trained on Historical Data
  (80/20 Train–Test Split)
- Target Probability >= 31%
- Targets 1060 customers
  with a Net LTV of $15,690
  and a 46% conversion rate!

WHY NOT GO BELOW 31%?

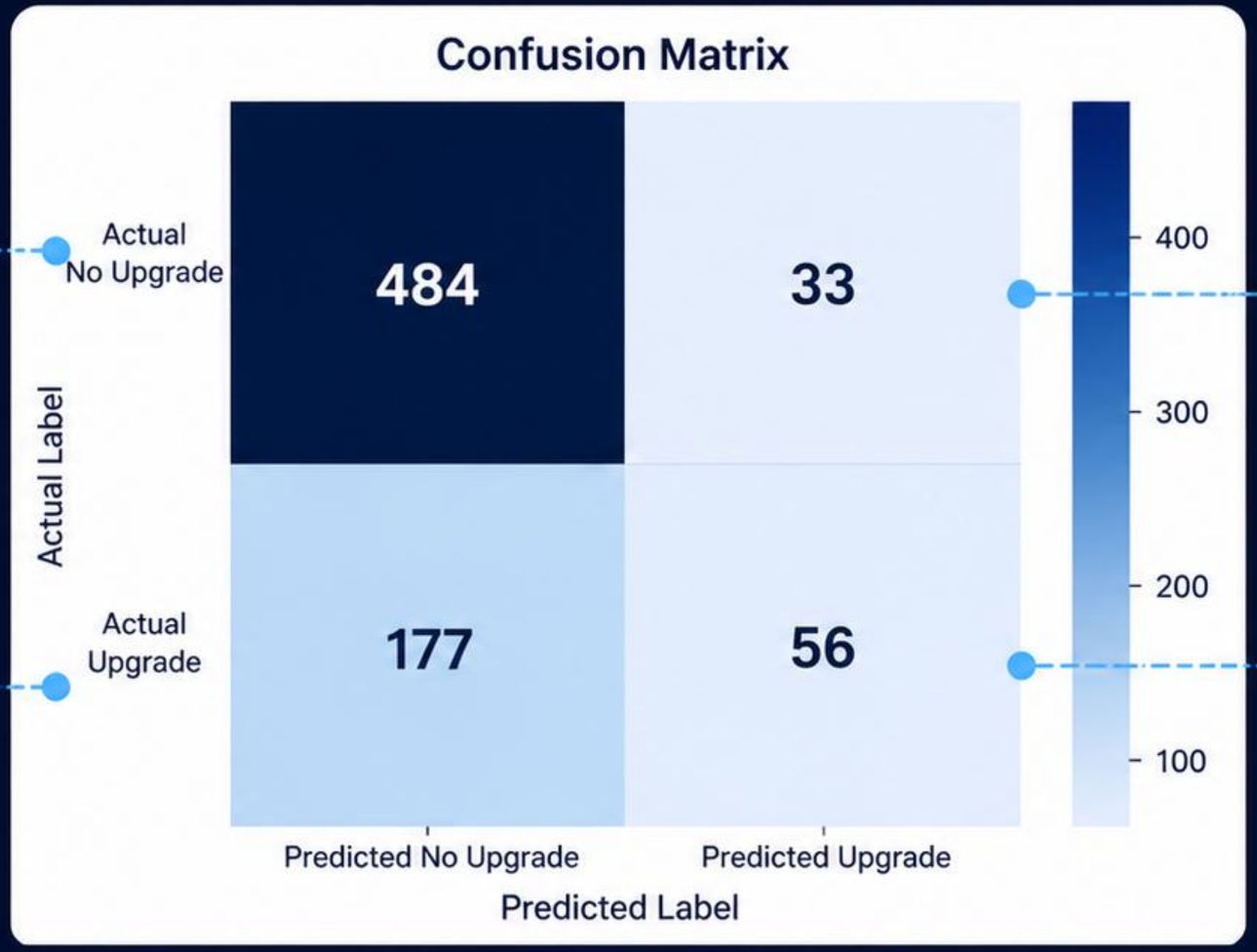


The lower you go,
the more
"False Positives"
you face,
and each of them
cost $55 each,
reducing Net LTV.

CONFUSION MATRIX

True Negative
Correctly predicted no upgrade.

False Negative:
Missed Revenue; No Loss



False Positives:
Cost $55 Each

True Positives:
Gain $96 Each

Key Takeaway: Maximize True Positives and True Negatives, while minimizing False Positives and False Negatives.

MODEL EVALUATION: WHY PRECISION MATTERS MOST

True Positives (TP)

+$96 revenue per correctly identified upgrader. These are subscribers we target who accept the Premium offer.

False Positives (FP)

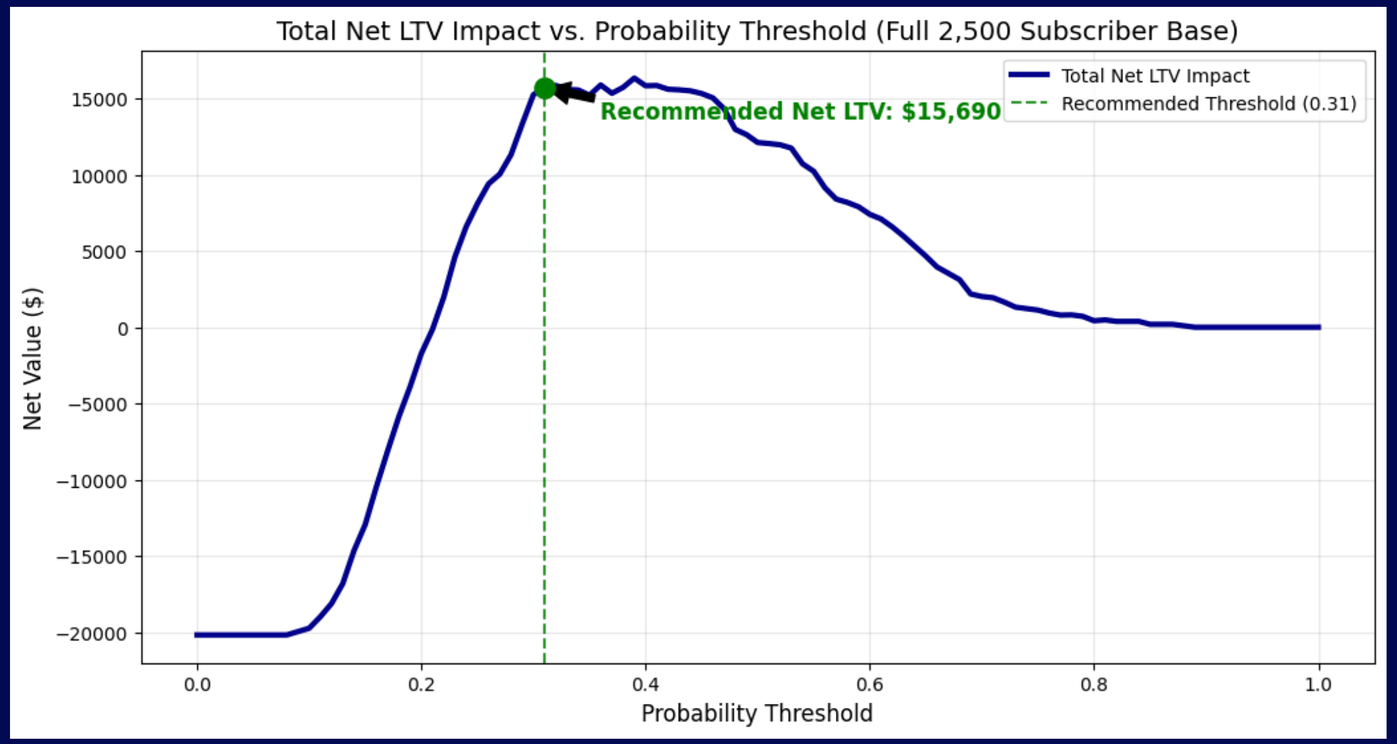
~$55 LTV loss per incorrectly targeted subscriber who churns after receiving an unwanted offer.

Key Metrics Focus

Precision is critical: TP/(TP+FP). High precision minimizes costly churn while maximizing net value.

NET VALUE OPTIMIZATION & THRESHOLD SIMULATION





Net Value = (TP × $96) – (FP × $55)

Low threshold → high recall, more FP

High threshold → high precision, more FN

OPTIMAL THRESHOLD IDENTIFIED

Optimal targeting threshold at predicted probability = 0.31 maximizing net value while balancing precision and recall for the StreamPlus Premium upgrade campaign.

	Threshold	True Positives	False Positives	True Negatives	False Negatives	Net Value	Precision	Recall	F1 Score
31	0.31	147	163	354	86	5147	0.474194	0.630901	0.541436
32	0.32	138	150	367	95	4998	0.479167	0.592275	0.529750
30	0.30	151	176	341	82	4816	0.461774	0.648069	0.539286
39	0.39	97	84	433	136	4692	0.535912	0.416309	0.468599
35	0.35	120	125	392	113	4645	0.489796	0.515021	0.502092
33	0.33	128	139	378	105	4643	0.479401	0.549356	0.512000
40	0.40	92	77	440	141	4597	0.544379	0.394850	0.457711
41	0.41	89	72	445	144	4584	0.552795	0.381974	0.451777
36	0.36	113	114	403	120	4578	0.497797	0.484979	0.491304
45	0.45	75	48	469	158	4560	0.609756	0.321888	0.421348

Note: that this optimal threshold is derived from a test dataset of 750 customers (30% of customers)

Target Audience

Targeting 1,060 subscribers representing 42.4% of the total subscriber base.

Conversion Rate Improvement

Projected 46.2% conversion rate vs. 11% historical untargeted.

Net LTV Impact

Projected $15,690 expected Net LTV impact from optimized targeting strategy.

RECOMMENDATIONS & EXPECTED IMPACT

Precision-Focused Targeting

Target only Basic subscribers with predicted upgrade probability ≥ 0.31 to maximize net value.

Eliminate Untargeted Campaigns

Stop broad untargeted campaigns immediately to avoid unnecessary churn and LTV loss.

Projected Outcomes

- Significant increase in campaign efficiency
- Stronger revenue conversion rates
- Substantial reduction in FP churn risk.

LIMITATIONS & FUTURE WORK

LIMITATIONS

Current model limited by **feature set** and **static nature**. (no causal inference, only correlations.).

Imbalanced classes affect model performance.

FUTURE ENHANCEMENTS

- Incorporating detailed **consumption patterns** and demographics
- Exploring **tree-based models** (Random Forest, XGBoost)
- Conducting controlled **A/B** experiments
- Developing **real-time scoring** with continuous monitoring.

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THANK YOU

